



Temporal attention to estimate continuous blood pressure using photoplethysmography signals

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Abstract

Continuous blood pressure (BP) monitoring plays a crucial role in preventative health care because it provides invaluable assessment of an individual's cardiovascular health. However, achieving accurate and non-invasive BP estimation remains a significant challenge. Motivated by the need for improved accuracy in BP estimation, this research proposes a novel deep-learning technique to estimate blood pressure from photoplethysmography (PPG) signals based on the temporal attention mechanism by selecting optimal features to improve the estimation accuracy. The proposed deep learning model consists of an embedded temporal attention employed to learn the temporal sequence and selects the optimal competitive features using bidirectional encoder and decoder. PPG features are extracted from traditional feature extraction techniques such as statistical, frequency, time domain-based analysis and chaotic features using two publicly available datasets MIMIC II and PPG-BP. The comparison of proposed model experimental results with the state-of-the-art models demonstrates the advancement of optimal feature selection technique for blood pressure measurement. The mean absolute error (MAE) and standard deviation SD for optimal feature extraction with the temporal attention mechanism model are 2.31 ± 3.16 mmHg and 1.13 ± 2.32 mmHg utilising MIMIC II data and 3.43 ± 3.08 mmHg and 3.29 ± 2.19 mmHg for PPG-BP dataset for SBP and DBP respectively. The proposed model results satisfied the standard requirements published by the British Hypertension Society (BHS) and US Association for the Advancement of Medical Instrumentation (AAMI) standards. The proposed solution can offer continuous BP estimation and can significantly enhance healthcare by helping individuals manage their cardiovascular health more effectively.

Keywords PPG · Datasets · Deep learning · Machine learning · Blood pressure · Features extraction

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1 Introduction

High blood pressure, also referred to as hypertension, is a leading risk factor for cardiovascular disease and a global health concern. One of the vital signs that helps medical professionals to assess a patient's cardiovascular health is blood pressure. Blood pressure measures the blood circulation pressure all over the body through the blood vessels. Typically, it is stated as systole, represented by systolic blood pressure, and diastole indicated as diastolic blood pressure [1]. The ideal range for SBP and DBP pressure is 90/60 and 120/80 mmHg, respectively [2]. There are several invasive and non-invasive blood pressure measurement methods. When necessary, invasive procedures are carried out by a medical professional in the critical care unit or during major surgical procedures. It is time-consuming and poses health hazards. Non-invasive methods are often employed due to

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their clinical value. There are several instruments available for non-invasive BP measurement. Most of these techniques allow for precise blood pressure monitoring without causing patients any discomfort. However, these methods use cuff-based blood pressure estimation devices that cannot be used for continuous estimation. Continuous BP monitoring is desirable in some situations when there is a risk of getting a stroke or heart-related disease. Researchers have been working on cuffless, non-invasive, continuous BP measuring methods to cope with this limitation.

In the past decade, numerous cuffless and non-invasive BP measurement approaches have been proposed as a substitute for traditional cuff-based and invasive BP devices. Two types of techniques are studied broadly for blood pressure estimation using pulse arrival time (PAT) and pulse transit time (PTT). These strategies, however, are not practically applicable since these techniques rely on specific physiological data of each person and necessitate the use of two sensors. This led researchers to consider and study PPG-based blood pressure estimation methods. Studies indicated that PPG sensor is an inexpensive, portable, easy-to-use technique that can be used for continuous BP monitoring. These techniques are becoming popular due to technological advancements, which means they can work through sensors. A PPG device uses a photodetector to illuminate the human skin and measures blood volume change by calculating the changes in light absorption [3, 4]. The PPG signal waveform resembles the blood pressure waveform and is synchronised with the blood pulse. The periodic PPG waveform provides systolic and diastolic peaks for each heartbeat cycle. The waveform depicts one turning point from systolic to diastolic, known as a dicrotic notch. It is easy to spot the systolic peak from the PPG waveform; however, diastolic and dicrotic notch points are challenging to detect for hypertensive or older individuals with other health conditions [5].

Monte-Moreno [6] measured SBP utilising the random forest (RF) machine learning model and PPG signal features. Later, Liu, Po [7] estimated BP using support vector regression (SVR) and extracted 21 PPG timescale-based features, which were initially proposed by Kurylyak, Lamonaca [8], and also extracted PPG's second derivative features (SDPPG). Previous studies have demonstrated that SDPPG may detect aortic stiffness and compliance, which are directly associated with blood pressure [9, 10]. This study suggested that combining the timescale and SDPPG features improved accuracy compared to other machine learning models. Yi, Jian [11] applied linear regression, Elastic network, LASSO, k-nearest neighbours (KNN), and classification and regression tree for BP estimation by extracting nine features. CART outperformed the other models; also, the authors specified that a small number of PPG features have

the potential to be as precise as the accuracy achieved with a more significant number of PPG features. However, they did not provide details about which features were included in this analysis. The authors [12] applied neural net, and various feature extraction techniques such as entropy-based, time-domain and frequency-domain methods for BP measurement were examined. The literature studies focused on feature extraction techniques and the results of these studies suggested strong evidence of the relationship between PPG features and blood pressure estimation.

Traditional machine learning models confront a significant accuracy degradation problem for continuous BP estimation. This problem arises as a result of their failure to model the temporal relationships inherent in the data, which need periodic calibration of patients [13]. Recently, Li, Harfiya [14] and [15] proposed an LSTM model for blood pressure estimation using seven features extracted from PPG and electrocardiogram (ECG). The results showed that this approach outperformed the methods presented in the literature. Senturk, Polat [16] compared different machine-learning methods for blood pressure estimation using PPG and ECG signals and the results complied with the AAMI standards. When examining a large dataset derived from diverse populations, linear models may struggle to effectively describe the BP and PPG relationship [17]. Traditional machine learning models, such as random forest and support vector machine (SVM) achieves high performance; however, if these models measure SBP and DBP independently, estimation accuracy will decrease because SBP is highly associated with DBP. Therefore, SBP and DBP should be estimated simultaneously by utilising a standard model architecture.

The literature studies suggested that the deep learning model can estimate blood pressure more accurately than traditional machine learning algorithms and showed a significant relationship between PPG features and blood pressure estimation. We conducted a benchmark study, and its results demonstrated that time-domain-based features obtained greater accuracy [18]. Furthermore, deep learning models such as Bi-LSTM or GRU using time-domain features performed better than all other models for blood pressure estimation [18]. Harfiya, Chang [19] recently studied the LSTM sequence-to-sequence model using PPG signals extracted from the MIMIC II dataset. The results satisfied the AAMI standards for BP estimation; though, the accuracy still needs improvement. Studies showed that the LSTM model is not always able to reliably memorise these lengthy input dependencies and the encoder-decoder network's performance gradually deteriorates, leading to the long-range dependency problem [20]. This problem can be solved by using an attention mechanism. It achieved surpassing performance with better results for sequential

analysis in many other domains related to machine learning and performed very well for natural language processing (NLP) like translation. Attention was first introduced for soft alignment of machine learning translation tasks in the NLP domain by [21]; they used the glimpse idea [22]. El-Hajj and Kyriacou [23] used an attention mechanism with a Bi-RNN structure for BP estimation using PPG signals. Two datasets were originated from one dataset (MIMIC II) for this research. The results of this study showed 2.58 ± 3.35 for SBP and 1.26 ± 1.63 (MAE $\pm$ SD) for DBP estimation. The outcomes of this study demonstrated the significance of attention mechanism for blood pressure estimation.

The above literature review illustrates that despite the numerous models developed for blood pressure estimation, several challenges remain unresolved. While they have proposed several methods for blood pressure estimation, however, they did not consider the temporal nature of Photoplethysmography signals. The literature review studies also did not focus on the most competitive features for Blood pressure estimation using photoplethysmography signals. These studies did not consider the temporal behaviour of PPG signals in the time domain analysis and detail analysis of traditional feature extraction methods, so the BP estimation methods need to be further investigated and improved. Since recurrent neural networks (RNNs) [24] have demonstrated strong performance in speech recognition and text generation, researchers have explored their application to time series prediction with some success. However, RNNs face a significant challenge: they are susceptible to gradient vanishing and exploding during training. To address these issues, advanced networks such as long short-term memory networks (LSTMs) [25] and gated recurrent unit networks (GRUs) [20] were developed. Although these models mitigate the gradient problems, their prediction accuracy remains suboptimal. So, several LSTM variants have been created, including stacked-LSTM [26] and bidirectional LSTM (BiLSTM) [27]. While these deep learning model variants show improved performance over standard LSTMs, there is a lack of research identifying which variants are best suited for temporal signals and specific time series [28]. Building on these LSTM models, encoder-decoder model, and temporal attention mechanism, this paper introduces a novel model for temporal time series prediction for PPG biosignals. We proposed a sequence-to-sequence encoder-decoder with a temporal attention mechanism model to select optimal features from PPG signals to increase the precision of continuous and cuffless BP monitoring using finger photoplethysmography.

The following research questions are of particular importance for this investigation. RQ1. How can the feature extraction and selection from PPG signals be improved for efficient accuracy? RQ2. How can we select the most

competitive features for blood pressure estimation? RQ3. What is the impact of feature selection on the performance of the proposed model prediction? The novelty of this study lies in its innovative application of a sequence-to-sequence encoder-decoder architecture with a temporal attention mechanism for PPG signals. To our best knowledge, this is the first study to consider PPG sequence-to-sequence analysis using temporal attention mechanism for optimal feature selection extracted from traditional techniques for blood pressure estimation. Unlike existing methods, this approach captures the temporal dynamics and long-term dependencies within PPG signals which makes it more effective for feature learning. The proposed method integrates traditional feature extraction techniques with deep learning-based temporal attention to data-driven optimization for PPG competitive feature selection. This addresses a critical gap in prior research, where feature selection often lacked rigorous optimization for predictive accuracy. The study proves its contributions as novel and impactful with superior performance benchmarking against state-of-the-art models. Specifically, these are the primary contributions of this study:

- Proposed a sequence-to-sequence encoder-decoder with temporal attention mechanism model to learn competitive optimal features from traditional feature extraction techniques to improve the blood pressure estimation performance using PPG signals using two datasets.
- Considered time-series based feature extraction from PPG signals to improve feature extraction effectiveness and efficiency to learn correlation of continuous feature and accurate blood pressure estimation.
- Combining features extracted from traditional features extracted techniques for the selection of most competitive feature with higher prediction ability to estimate blood pressure.
- Analysed and compared the proposed Bidirectional LSTM Encoder and Decoder with temporal attention mechanism model results with baseline experiments to analyse the performance.
- The proposed model results were evaluated using standard protocols and compared with related studies; the comparison with state-of-the-art models demonstrated the efficacy of competitive feature selection by proposed model.

This paper is organised as follows: Sect. 2 illustrates the method utilised for determining competent features by analysing PPG signals and estimate blood pressure, and this section presents datasets, pre-processing feature engineering, and the proposed methodology. Section 3 details the training proposed model and evaluation metrics. Section 4

provides detail discussion and results. Section 5 provides context for upcoming work while presenting the conclusion.

2 Materials and methods

2.1 Experimental data and pre-processing

Two publicly available datasets were used for this study: the Multiparameter Intelligent Monitoring in Intensive Care dataset (MIMIC-II) [29], and the PPG-BP dataset [30]. The MIMIC II dataset provides health-related data from thousands of patients admitted to the critical unit of a hospital. This dataset is widely used for BP estimation methods and models since it contains PPG signals and references BP signals. It consists of physiological signals such as blood pressure, pulse rate, electrocardiogram, and breathing rate and sampled at 125 Hz. PPG and blood pressure signals are segmented into 30-second windows, each of which displays consistent blood pressure values. It also includes patient data such as age, heart rate, weight, height, body mass index (BMI), systolic blood pressure, diastolic blood pressure and PPG biosignals. The PPG-BP dataset records are from 219 patients aged 21 to 86. This dataset contains 15.5% stage 1 hypertension, 9.2% stage 2 hypertension, 38.8% prehypertension, and 36.5% normotension patients. Each patient's three PPG signal segments were recorded with a sampling frequency of 1 kHz. The signals were recorded for 2.1s with sample lengths of 2100.

The feature extraction from biosignals is crucial for constructing a trustworthy and generalised approach. Many PPG and BP signals have distorted and irregular parts in the signal databases. To efficiently extract PPG characteristics, we used multiple methods. In the first step of this process, we aligned the PPG and BP signals after removing any irregular signal segments. It is possible that the base of these signal acquisitions will shift as a result of the patient moving around while the signal is being recorded.

These movement artefacts and noise can have an effect on the signals depending on the properties of the sensor, such as the measurement point of the patient, the movement of the patient, and the ambient light. Denoising the signals and removing these movement artefacts required the application of a signal processing filter known as the Butterworth. The passband frequency response is supposed to be as mathematically flat as can be achieved with this design. In some circles, it is referred to as a flat maximum magnitude filter [31]. The Butterworth frequency filter with the cut-off frequencies (0.5 to 30 Hz) and the order of 2 is used with a lowpass filter type to denoise the required signals for both datasets. In addition, the min-max normalisation technique was used on the PPG biosignals in order to bring them into a range that was between 0 and 1 utilising the PPG signals. This range was achieved by applying the min-max normalisation method. These biosignals were further divided into the same time frames and used as input datasets for feature extraction (See Fig. 1 and 2).

2.2 Traditional feature extraction techniques

Most of the research on PPG has used a features engineering approach, with features extracted from signal waves, second derivative, and other methods given in Table 1. To address concerns about trust and reliability, we planned to extract features traditionally initially and then distinguish the significance of different features depending on the given new sequence of inputs at a given time. In addition, a deep learning model will learn the signal pattern for optimal feature selection.

Frequency-based features are significant for biosignals analysis. We used the multitaper method, an extension of spectral representation [32]. This method is used to deal with the limitations of the classic spectral method. A more accurate spectrum estimation can be obtained by utilising a combination of low variance and high frequency than with the traditional and Welch's periodograms that can be seen

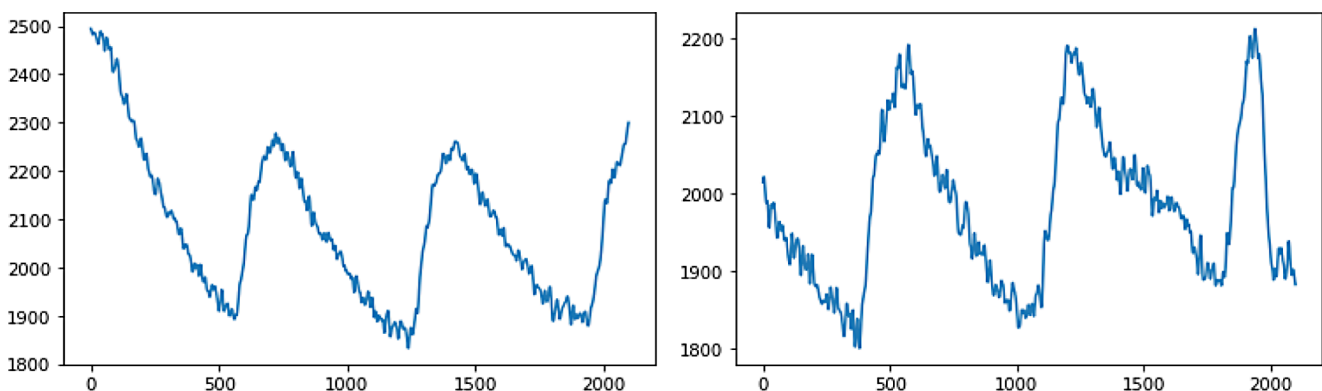


Fig. 1 Inappropriate PPG signals. Some significant features, such as a dicrotic notch, cannot be extracted from some PPG signals; however, they can be approximated from some signals

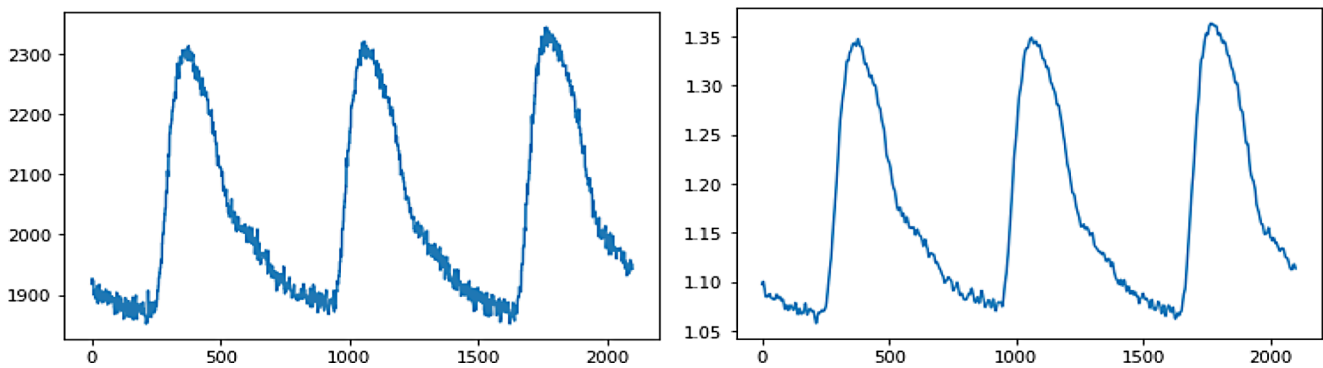


Fig. 2 PPG signal before and after pre-processing using Butterworth lowpass filter to denoise the signals

Table 1 Summary of feature extraction techniques used in literature

Feature extraction techniques	Literature review
Frequency-based features	Mahbub and Rabbani [39], Senturk, Polat [16], Maqsood, Xu [18], Nishan, Raju [40]
Statistical features	Chowdhury, Alzoubi [41], Elgendi [42], Maqsood, Xu [18]
Time domain features	Nishan, Raju [40], Teng and Zhang [43], Suzuki and Oguri [44], Lamonaca, Barbe [45], Suzuki and Ryu [46], Kachuee, Kiani [47], Fujita, Suzuki [48], Bramwell and Hill [49] Nichols, O'Rourke [50], Allen and Murray [51], El Hajj and Kyriacou [52], Mukkamala, Hahn [53], Maqsood, Xu [18], Wagenseil and Mecham [54], Sharma, Barbosa [55], McCombie, Reisner [56]

in Eq. 1. Moreover, x_t may consist of multiple elements that appear recurrently provided in Eq. 2. Equation 3 calculates the $dZ(f)$ value which is a discrete orthogonal increment process and cannot be zero (0). Further, Eq. 10 calculates the second central moment of $dZ(f)$.

$$x_t = \int_{1/2}^{1/2} e^{-i\omega t} dz(t) \quad (1)$$

$$\begin{aligned} x_t &= \sum_j C_j \cos(\omega_j + \varphi_j) + \xi_t \\ &= \sum_j \mu_j e^{-i\omega t} + \mu_j^* e^{-i\omega t} + \xi_t \end{aligned} \quad (2)$$

$$E\{dZ(f)\} = \sum_j \mu_j \delta(f - f_j) df, \quad (3)$$

$$E\{|dZ(f) - E\{dZ(f)\}|^2\} = S(f) df \quad (4)$$

When determining whether periodic components are noteworthy for MTM spectral estimation, the multiwindow approach offers a functional and uncomplicated likelihood ratio test. This method makes use of a large number of data windows, which are also referred to as “Slepian sequences” and “discrete prolate spheroidal sequences”. The following Eq. 5 is a definition of each of these terms. Where W

is the bandwidth of spectral, N is the sample number of the PPG signal, and λ_k represents the Slepian sequences $v_n^{(k)}(N, W)$ [33]. We can calculate Slepian functions using Fourier transformation as given in Eq. 6.

$$\lambda_k v_n^{(k)}(N, W) = \sum_{m=0}^{N-1} \frac{\sin\{2\pi W(n-m)\}}{\pi(n-m)} v_m^{(k)}(N, W), \quad (5)$$

$$v_k(f) = \sum_{n=0}^{N-1} v_n^{(k)}(N, W) e^{-i2\pi f n}. \quad (6)$$

$$y_k(f) = \sum_{t=0}^{N-1} x_t v_i^{(k)}(N, W) e^{-i2\pi f t}. \quad (7)$$

The first thing that the MTM does is compute the expansion coefficients, also known as eigen coefficients, of the input X_t . This may be seen in the following Eq. 7. The value that is projected to be returned by $y_k(f)$ can be computed by combining the formulas that have been presented previously. One method for establishing this is by reducing the amount of residual local squared error, which happens when f equals f_0 in the calculation. The squared error can be defined using the equation that follows (10). The findings can be presented in the form of an Eq. (11). The Eq. (12) is utilised to calculate the spectrum continuous sector. However, due to the presence of a significant line component with the frequency f_0 , it is necessary to reconstruct the spectrum in accordance with the Eq. (4).

$$E\{y_k(f)\} = \mu V_k(f - f_0) + \mu^* V_k(f + f_0) \quad (8)$$

$$\begin{aligned} E\{y_k(f_0)\} &= \mu V_k(0) + \mu^* V_k(2f_0) \\ &\approx \mu V_k(0); \text{ where } f = f_0 \end{aligned} \quad (9)$$

.

$$e^2(\mu, f) = \sum_{k=0}^{N-1} |y_k(f) - \mu(f) V_k(0)|^2. \quad (10)$$

$$\hat{\mu}(f) = \frac{\sum_{k=0}^{K-1} V_k^*(0) y_k(f)}{\sum_{k=0}^{K-1} |V_k(0)|^2}. \quad (11)$$

$$\hat{S}(f) = \frac{1}{K} \sum_{k=0}^{K-1} |y_k(f)|^2. \quad (12)$$

$$\hat{S}_r(f) = \frac{1}{K} \sum_{k=0}^{K-1} |y_k(f) - \hat{\mu}(f_0) V_k(f - f_0)|^2. \quad (13)$$

Furthermore, **statistical features** extracted from PPG bio-signals for blood pressure estimation. In this section, we discuss the statistical features we extracted. These features are Skewness, Kurtosis, Perfusion and maximum, minimum and mean absolute deviation. Equation (14) is provided to calculate skewness where σ and $\hat{\mu}_x$ show the experimental evaluation of the average mean (x_i) and standard deviation, and N represents the sample number from the PPG signal. Perfusion is a standard technique to determine the quality of signals. This may be computed by the static tissue blood flow or by taking the pulsatile to non-pulsatile ratio. In other words, this demonstrates the disparity concerning the amount of light transmitted and absorbed by the pulse and the skin tissue. That is further calculated by Eq. (15), where $\bar{x}$ is the mean of x PPG signal, and y is the filtered PPG signal.

$$S = 1/N \sum_{i=1}^N [x_i - \hat{\mu}_x / \sigma]^3 \quad (14)$$

$$P = [(y_{max} - y_{min}) / |\bar{x}|] \times 100 \quad (15)$$

Kurtosis identifies the observed data distribution, such as peakedness and heavy tail or flat distribution, or light tail compared to the normal distribution. Equation (16) can be used to calculate kurtosis where N shows the number of PPG signals samples, σ represents the average mean, and $\hat{\mu}_x$ presents the standard deviation. The maximum,

minimum, and mean absolute deviation features can be calculated using Eqs. (17–19), where x represents the signal.

$$K = 1/N \sum_{i=1}^N [x_i - \hat{\mu}_x / \sigma]^4 \quad (16)$$

$$\text{maximum}(x) = \max(x_i). \quad (17)$$

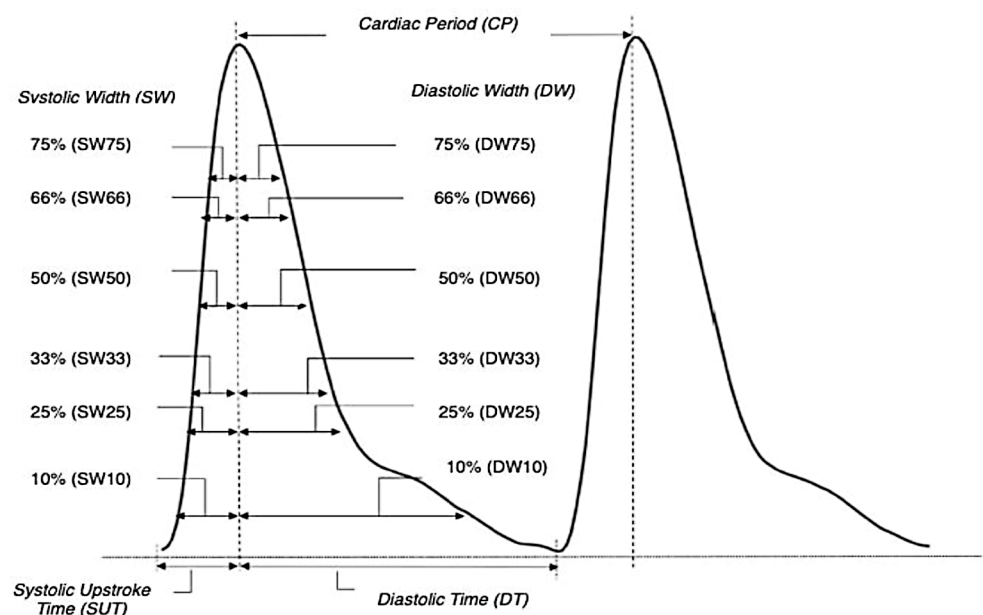
$$\text{minimum}(x) = \min(x_i) \quad (18)$$

$$\text{MeanAbs}(x) = \frac{1}{N} \sum_{i=1}^N |x_i - \bar{x}| \quad (19)$$

Moreover, we extracted **time domain features** which is the amplitude and duration of each cardiovascular cycle element define the PPG waveform. Due to movement artefacts occurring during PPG recording, the pulse height may fluctuate significantly, making blood pressure computation difficult. Conversely, oximeters and other sensor devices that generate the PPG signals gather frequencies at different rates according to the length of time between each heartbeat [34, 35]. Distinct values were retrieved for the systole area of the signal (i.e., the gap amid the minimum and maximum marks). For the diastolic component (the wait concerning the maximal and the subsequent minimal point). We adopt 21 metrics extracted from the PPG signal, containing the time between heartbeats, systolic and diastolic readings, and their ratios that are shown in Fig. 3.

Moreover, we extracted chaotic features which was found that a complex feature was able to provide a complete explanation of time-series signals. Multidimensionality or segmenting the signals into identical lengths are two ways to identify a signal. Complex features were used to provide a complete explanation of time-series signals. The signals can

Fig. 3 Characteristics derived from each cycle of the PPG waveform are in the time domain. These parameters were derived using signal amplitude, such as systolic or diastolic, throughout a range of waveform widths



be recognised by their unique characteristics, such as their multidimensionality or the ability to be divided into uniform lengths. Senturk, Polat [16] features of fractal dimensions were extracted by using Higuchi [36] and Katz [37]. Senturk, Polat [16] employed entropy as a form of chaotic feature extraction, and we were able to identify disruptions in ECG and PPG signals. By utilising fractal dimensioning, which involves subdividing the signal into sub-signals of varying sizes, the dimensions can be subtracted. Higuchi fractal dimension [36] features can be calculated as follows: With this method, you can generate k new time series with varying resolutions.

X_k for $m = 1, \dots, k$.

$$X_k^m = x(m), x(m+k), x(m+2k), \dots, x(m + \lfloor \frac{N-m}{k} \rfloor k) \quad (20)$$

The length $l(k)$ of curve X_k^m is calculated as given below:

$$l(k) = \frac{\left(\sum_{i=1}^{\lfloor \frac{N-m}{k} \rfloor} |x(m+ik) - x(m+(i-1)k)| \right) (N-1)}{\left(\frac{N-m}{k} \right) k} \quad (21)$$

The typical range for the length of each k from l to k_{max} is determined by taking the average of the k lengths. Katz fractal dimension [37] the algorithm can be derived in a straightforward manner from the time series using Eq. 22. The entire duration of the PPG or ECG biosignal is indicated by this notation “L,” while “d” represents the Euclidean distance between the starting point and the point that specifies the additional distance concerning the starting point. In addition, “a” stands for the typical number of steps in a series.

$$FD_{Katz-norm} = \frac{\log_{10} \left(\frac{L}{a} \right)}{\log_{10} \left(\frac{d}{a} \right)} = \frac{\log_{10} (n)}{\log_{10} \left(\frac{d}{L} \right) + \log_{10} (n)} \quad (22)$$

Entropy: Predicting blood pressure is tough because it is a non-linear, complex signal. In order to decipher complicated signals, entropy is a useful tool. Obtaining complex numbers can yield a wealth of information. Complex health signals can be quantified by entropy [38]. These signals can be used to determine their Shannon entropy, approximate entropy, and permutation entropy [16]. Simple entropy for the Signal X for $i = 1, \dots, N-1$ as given below:

$$Entropy = \sum_{i=0}^{N=1} p_i \log(1/p_i) \quad (23)$$

2.3 Model building

In this research, we designed a model to learn sequence-to-sequence competitive features from input PPG signals and estimate blood pressure continuously and effectively. The proposed model network structure is shown in Fig. 4. Optimal feature extraction with temporal attention mechanism architecture comprises an input layer, bi-directional LSTM encoder, temporal attention layer and LSTM decoder layer. The input layer receives the feature vectors extracted from the PPG signals. These feature vectors represent various physiological parameters or signal characteristics that are relevant for blood pressure estimation. Each feature vector corresponds to a specific time stamp in the input sequence. Several features have been extracted using traditional features extraction techniques and fed into the architecture as a series of input $X_t = \{X_i^n | i = 1, 2, \dots, p; n = 1, 2, \dots, k\} \in R^{p \times k}$, where k presents the number of features and p presents the total number of samples on each time stamp $t \in [T_i]$. These feature vectors were divided into two categories; one is observed input series denoted as $x_1, x_2, \dots, x_n$, and the other one is predicted denoted as $p_1, p_2, \dots, p_n$.

An embedding layer is used to flatten the input layer from both input series to facilitate convergence. These features were further provided to Bi-LSTM to assign the weights for the hidden states to produce different feature vectors, h^1, h^t, h^n . The Bi-LSTM is capable of capturing temporal dependencies in both forward and backward directions, allowing it to effectively encode sequential information from the input PPG signals. This layer generates hidden states h^1, h^t, h^n at each time step, which encapsulate the learned representations of the input sequence. There is a problem with the LSTM encoder in that it holds hidden information of the previous state into a fixed-length context vector denoted by C_T . Thus, increasing input data length decreases the architecture's prediction ability [21]. We used temporal attention to address this issue by providing an interface between the encoder and decoder. At each time step t , the encoder input $(x_1, x_2, \dots, x_n)$ with its internal hidden states h , LSTM takes it as follows:

$$\begin{aligned} \vec{h}_t &= LSTM(x_t, \vec{h}_{t-1}); \overleftarrow{h}_t = LSTM(x_t, \overleftarrow{h}_{t-1}); \\ h_t &= \vec{h}_t \oplus \overleftarrow{h}_t \end{aligned} \quad (24)$$

Where the direction of arrows indicates the processing direction of Bi-LSTM, the feature vectors from the Bi-LSTM encoder are further wired into the temporal attention mechanism. The attention mechanism selects the high-source information by generating the weighted sum of hidden states received from the Bi-LSTM encoder and sends

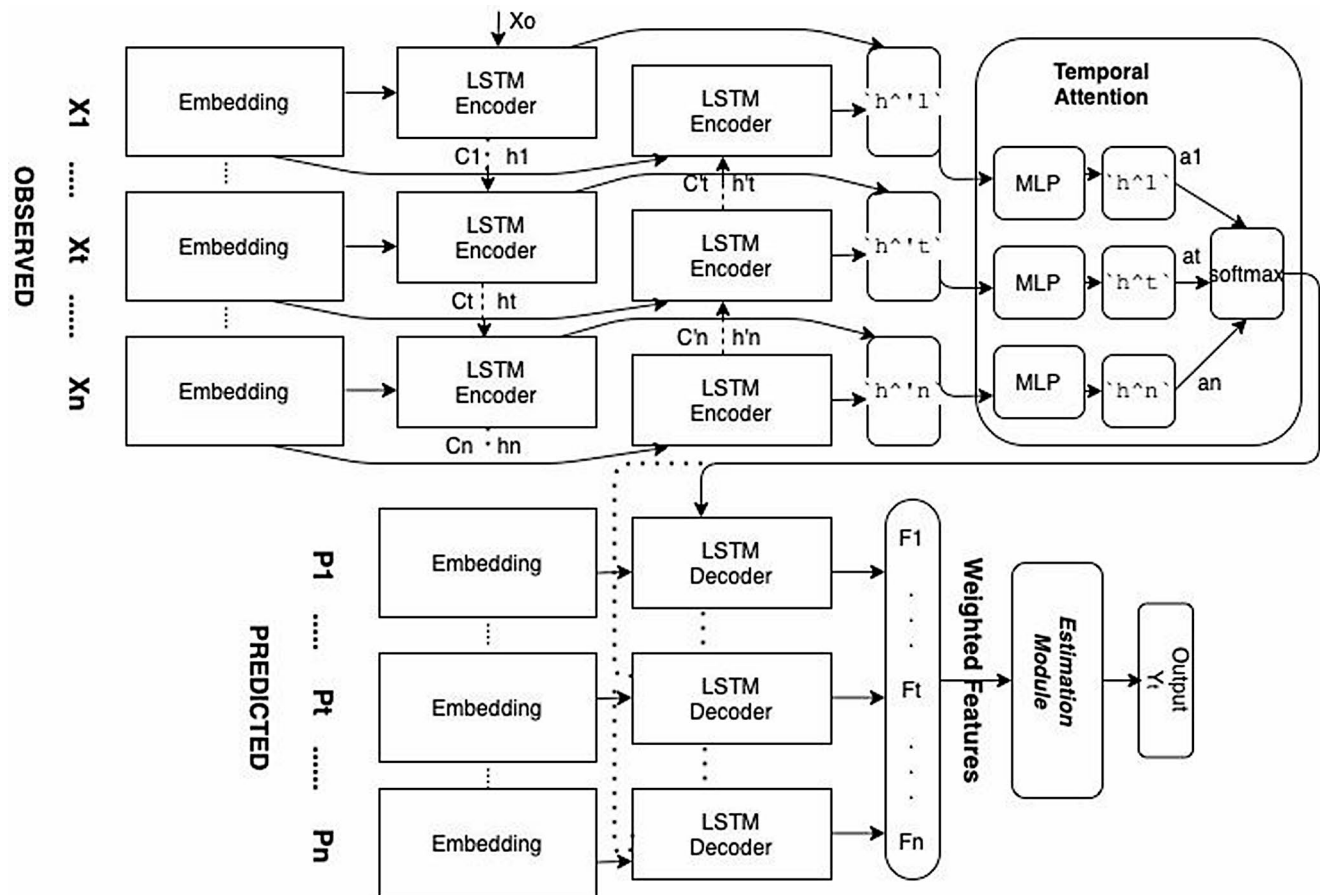


Fig. 4 Shows the proposed architecture of Optimal Feature Extraction with Temporal Attention Mechanism. There is a series of features input, $X_t = \{X_i^n | i = 1, 2, \dots, p; n = 1, 2, \dots, k\} \in R^{p \times k}$, where k presents the number of features and $p \{P_1, \dots, P_t, \dots, P_n\}$ presents the

total number of samples on each time stamp $t \in [T_i]$. Two categories are created from the input features, of which one is observed, and the other can be predicted. The encoder applies Bidirectional LSTM units to capture the significance of weights from historical feature series

this information to the LSTM decoder. At each time step t , the temporal attention mechanism dynamically weighs the importance of each hidden state h^n based on its relevance to the current context. This dynamic weighting allows the model to focus more on informative hidden states while downplaying the significance of less important features. The output of the temporal attention mechanism serves as the input to the LSTM decoder layer. This layer receives the dynamically weighted representations of the input sequence and decodes them to produce the competitive and optimal feature values. The decoder layer utilizes LSTM cells to decode the context vector and generate the output sequence. The output sequence represents the weighted features at each time step, providing continuous and sequential predictions. LSTM decoder generates the target sequence by decoding the weights information received from the attention mechanism. The target feature set is generated with competent features for blood pressure estimation. The optimal feature vector is then provided to the estimation module for SBP and DBP estimation.

2.4 Bidirectional encoder–decoder

We used an LSTM-based encoder-decoder as the foundation of our proposed sequence-to-sequence architecture analysis. In the context of long-range modelling dependencies, the specialised RNN structure known as LSTM has been demonstrated to be stable and robust [57–59]. We used an LSTM encoder-decoder. However, other models may also be employed for comparability with widely used sequence-to-sequence baselines. Additionally, it acts as an alternative to conventional positional encoding by offering the inputs' time ordering with the proper inductive bias. Simple LSTM captures the information related to the previous input state by the hidden state that has already been processed. Hence, the model can preserve the past information from the input in a unidirectional connection. However, the model unable to learn the future state in the same sequence. To improve the accuracy and learn additional information about the input states such as past, present and future, a bidirectional RNN connection can be used for training the neural network using

the sequence input for the previous and upcoming information [60]. Two interconnected layers operate in the same and opposite directions as the data sequence [61]. In particular applications, such as phoneme classification, Bi-LSTM proved more effective than unidirectional LSTM [62]. This allowed the combination of forwards and reversed hidden layers for Bi-LSTM calculation.

In this proposed model, we used Bi-LSTM as the encoder and LSTM as the decoder module that generates the target values on each time step by considering the previous temporal context. Encoder LSTM attempts to capture all input feature $(p_1, p_2, \dots, p_n | X_1, X_2, \dots, X_T)$ information and store it in its final internal states where X presents the observed and p presents the predicted features. The decoder component receives the internal states of the encoder and attempts to create the target sequence using them. To be more specific, the Bi-LSTM encoder encodes the input source data into a context C_T generation that is a temporal context of input data and a fixed vector. LSTM decoder is responsible for generating the target sequence by decoding the context C_T . The probability of this target sequence can be computed as follows:

$$P(p_1, p_2, \dots, p_n | X_1, X_2, \dots, X_T) = \prod_{t=1}^n P(p_t | C_T, f_1, f_2, \dots, f_{t-1}) \quad (25)$$

There are observed feature vector input is provided using separate LSTM encoders to produce different feature vectors, h^1, h^t, h^n . These feature vectors from the Bi-LSTM encoder are further wired into the temporal attention mechanism. The hidden and cell states of a Bi-LSTM network are initialised to zero. The latent representation is created by joining the hidden states when the recurrent method is used to process data $h_n \rightarrow$ and $h_n \leftarrow$ from both directions. The embedding layer will be utilised to flatten the inputs at time t . The encoder utilises bidirectional LSTM units to determine the significance of weights from historical feature series. The weight vector for attention is learned at each t time step from hidden states of bi-directional LSTM and serves as the input for the decoder LSTM for generating the attention weights to get optimal features. LSTM utilises the gates to control the information passing through each time step and perform temporal dependencies with time series. The sequence of hidden states output at a different time can be used for multi-step predictions. In this scenario, we got two hidden states at every time step that further is concatenated into one by applying $h_t = [h_t(\text{forward}); h_t(\text{backward})]$.

$$\vec{h}_t^n = \overrightarrow{LSTM}(x_t, \vec{h}_{t-1}) \quad (26)$$

$$\overleftarrow{h}_t^n = \overleftarrow{LSTM}(x_t, \overleftarrow{h}_{t+1}) \quad (27)$$

For biosignals, points of significance in time series data are frequently identified in relation to their surrounding environmental confounder's values, such as noise, anomalies, patient movement and cyclical patterns. In attention-based architecture, the selection of features from such pattern information can result in performance enhancements. This attention vector weight is then applied to the decoder as an input which contains weights for predicted input series and can calculate optimal feature selection. Further, these features forward to the estimation module for blood pressure estimation denoted by Y_t .

2.5 Temporal attention

Let the dataset is presented by a series of feature inputs matrix $X_t = \{X_i^n | i = 1, 2, \dots, p; n = 1, 2, \dots, k\} \in R^{p \times k}$, where k presents the number of features and p presents the total number of samples on each time stamp $t \in [T_i]$. Each row of the record shows a feature vector. The attention module computes the weights for all features. As depicted in Fig. 4, the attention module is at the heart of the entire framework. The estimation module seeks the best correlation between the weighted features and the final SBP and DBP prediction results. The attention and estimation modules work together to create the correlation that best describes the degree of relevance of the target and features. In this study, the temporal attention model learns the weights of all hidden states of LSTM. Further, implementing a multilayer perceptron makes a dynamic temporal relation with the hidden stage that can be changed over time to get better features. Attention weights are normalised from all hidden states using a soft-max function. The context vector (q_t) is used to create the temporal attention using LSTM hidden states and the calculated attention weights. The context vector calculates the alignment scores ($\alpha_{t,j}$) and holds all output hidden state information. Given an encoder h_j as the hidden state at time-step j , and decoder p_t as the hidden state at time-step t .

$$q_t = \sum_{j=1}^T \alpha_{t,j} h_j \quad (28)$$

$$\alpha_{t,j} = \frac{\exp(\text{score}(p_{t-1}, h_j))}{\sum_{j=1}^T \exp(\text{score}(p_{t-1}, h_j))} 1 > \alpha_i^t > 0 \quad (29)$$

$$p_t = f(p_{t-1}, y_{t-1}, q_t) \quad (30)$$

2.6 Estimation Module

The optimal weighted features F_n provided as an input to the estimation module. The predicted input p_t and temporal attention are used to estimate optimal weighted features F_n . This process works for each time step t and keeps learning the optimal features while estimating SBP and DBP. We used a deep neural network (DNN) to estimate SBP and DBP using optimal weighted features. A deep neural network (DNN) is an advanced network capable of learning more complicated and abstract properties than traditional neural networks [63]. DNN consists of an input layer (i), hidden layers n and an output layer o . DNN is efficient for processing numerical data [64]. In the estimation module, we received a set of optimal weighted features $F_1, F_2, \dots, F_n$ after attention and bi-directional LSTM weight calculations as an input vector F_n where n is a number of features given to hidden layer h_n . The hidden layer learns the optimal features and provides the output y_t , which are the values of SBP and DBP.

3 Model training

We use Google collaboratory¹ (Colab pro) to perform the experiments for our proposed model using two datasets. Further, we use Python 3.7² Programming language and built-in libraries to support machine learning algorithms such as sci-kit-learn, pandas, TensorFlow, NumPy, Matplotlib, Keras, math, etc. The parameters optimisation includes the number of hidden layers, learning rate, batch size and the number of epochs. The proposed model is optimised using the Adaptive Moment Estimation (Adam) [65]. We extracted optimal features for BP estimation, which is the regression problem, using the mean square error (MSE) as the loss function, a typical objective function for regression problems. During the training of the neural model, optimisation is one of the crucial steps used to change and set the neural network parameters, such as learning rate and weights, to reduce the error. The error used to identify the loss value is known as the cost function. The lowest point of error is known as global or local minima. For SBP and DBP estimation values, we use the MSE cost function.

Table 2 Hyperparameters setting for the experimental setup

Encoder hidden units	Decoder hidden units	Loss function	Optimiser	Learning rate	Attention activation function
256	512	MSE	Adam	10^{-3}	SoftMax

¹ <https://colab.research.google.com/>.

² <https://www.python.org/>.

$$MSE = \frac{1}{n} \sum_{i=1}^n (Y_i - \hat{Y}_i)^2 \quad (31)$$

It can be seen from Eq. 36 that Y_i presents the target value, $\hat{Y}_i$ shows the estimated value, and n is the number of data points. The cost function (MSE value) depicts the model performance, and the decline in MSE value increases the model's performance. The learning rate is set to 10^{-3} with a batch size of 64. The number of epochs was set to 300 with the early stopping method as the cost function began to converge. The longer training time might result in overfitting, leading the model to learn the irrelevant noisy information from the dataset and drop the performance [66]. To avoid this, we used dropout to reduce the model complexity set to 0.2. We used 256 hidden layer units for the Bi-LSTM encoder and 512 for the LSTM decoder to achieve optimal outcomes. For estimation module DNN implementation, we used three hidden layers with each 256 nodes and a linear activation function for one output layer for BP estimation. We use the sum function to concatenate the bi-LSTM weights achieved from both directions. Additionally, we used 5-fold cross-validation to thoroughly evaluate the robustness and generalizability of our model. In this approach, the dataset was divided into five equal parts. Each part took a turn as the validation set while the remaining four were used for training. This process was repeated five times, allowing us to average the performance metrics and evaluate the model across different subsets of data. This approach helped minimize the risk of overfitting and reinforced the reliability of our results. We found that the Adam optimiser performed better than the gradient decent optimiser during the optimisation process. Table 2 provides the optimal parameters for the experimental setup.

3.1 Evaluation metrics

Literature showed that mean absolute error and standard deviation are the widely used evaluation metrics in this research domain. Mean absolute error can be calculated by taking the average of the absolute differences between the actual and expected values. An average residual can be calculated by taking the average of all the residuals from the dataset, where $\hat{y}$ shows the expected value of y and $\bar{y}$ is the mean given in the Eq. (32). The second evaluation metric is the standard deviation (SD), a statistic expressing the degree to which a data group varies or disperses. It measures how closely related the values in a collection are to the anticipated values. However, a high standard deviation indicates a wider range of values. The standard deviation of expected values (SD) assists in comprehending the dispersion of projected values in a range of models. Mean absolute error

and standard deviation can be calculated using the formula shown below:

$$MAE = \frac{1}{N} \sum_{i=1}^N |y_i - \hat{y}| \quad (32)$$

$$SD = \sqrt{\frac{\sum (x - \bar{x})^2}{N - 1}} \quad (33)$$

3.2 Standarding Protocols

There are two widely used standard protocols for blood pressure estimation validation and testing. These were developed by the Association for the Advancement of Medical Instrumentation (AAMI) [67] and the British Hypertension Society (BHS) [68]. Both standards require three readings of blood pressure for eighty-five subjects to test the device's validity [69]. The standards set by AAMI are as follows:

- The golden values and predicted value mean absolute difference must be less than 5 mmHg.
- The standard deviation error should be less than 8 mmHg.

The standards set by the BHS are as follows [69]:

- The results from the blood pressure device should be within 5mmHg for at least 50% of experiments.
- Grade B indicates that 75% of the outcomes should be within 10 mm Hg.

4 Results and discussion

The significance of PPG signals and advanced machine learning techniques leads researchers to find optimal solutions for BP estimation. This new sequence-to-sequence Bi-LSTM model with temporal attention has several key advantages that make it a promising approach to blood pressure estimation. It is able to capture the long-range dependencies in sequential data and therefore is able to properly examine the temporal relationships within PPG signals in order to make more accurate predictions. The attention mechanism improves this capability in addition to that, it is able to focus on the most relevant features at each time step, thus the model is able to concentrate on the important signal features and suppress less significant or noisy parts. The proposed sequence-to-sequence Bi-LSTM model with temporal attention, while innovative, has its limitations. Its effectiveness depends on the quality of PPG signals, and despite applying the Butterworth filter, noise and artifacts

from patient movement or environmental conditions can still affect accuracy. The chosen band-pass filter range (0.5 to 30 Hz), while useful for denoising, may inadvertently exclude some clinically significant frequency components, which could limit the model's predictive capabilities. Additionally, the computational demands of the Bi-LSTM encoder-decoder architecture with attention mechanisms result in longer training times and higher resource requirements, making it challenging to implement in real-time or resource-limited environments. The model also risks overfitting, particularly with smaller or less diverse datasets, raising concerns about its generalizability to other datasets or clinical scenarios. Furthermore, it struggles with low signal-to-noise ratio (SNR) cases, such as signals from elderly or unhealthy individuals, where critical features like diastolic or dicrotic notches are harder to detect, potentially reducing accuracy. These challenges underscore the need for further refinement to enhance the model's robustness and ensure its broader applicability in clinical settings.

Harfiya, Chang [19] recently studied the LSTM sequence-to-sequence model to estimate blood pressure using PPG biosignals obtained from the MIMIC II dataset. The results satisfied AAMI standards to estimate blood pressure; however, the accuracy still needs improvement. Studies showed that the LSTM model does not always memorise the long input dependencies, and the encoder-decoder network's performance gradually deteriorates, leading to the long-range dependency problem [20]. This problem can be solved by using an attention mechanism. As the sequence-to-sequence encoder-decoder with attention mechanism has produced significant results in other domains, we consider it a practical approach to provide a continuous, cuffless, and non-invasive solution for BP estimation. To accomplish state-of-the-art performance, we use Bi-LSTM encoder and decoder with temporal attention mechanisms to extract the optimal features using signal-to-signal analysis. At each time step, the attention mechanism retains the entire input hidden states sequence and prioritises the most relevant input source. It achieved surpassing performance with better results for sequential analysis in many other domains related to machine learning and performed very well for NLP domains like translation [21, 22].

The features have been extracted using different feature extraction techniques (e.g., frequency-based features, chaotic features, statistical features extraction and time-domain features). The feature set represents a complex number of mixed values. These features may have varying effects on blood pressure estimation; some features can perform better but can be neglected. On the other hand, some features from the dataset contain less content to be predicted. In this situation, the attention mechanism automatically focuses on the most significant hidden states. The attention mechanism

assigns high weights to valuable hidden states containing competitive features for blood pressure estimation. The attention mechanism performs efficiently using the attention layer and provides the weighted sum of the Bi-LSTM hidden layer into the attention vector. This research study proposes improving biosensor signals' performance in information extraction, thus, better predicting cardiovascular diseases. Moreover, the biosignals data contain some signal pattern anomalies that occur while recording these signals, such as environmental artefacts. These signals might have notable features that can be extracted, such as it is easy to extract systolic blood pressure from the PPG waveform peak; however, difficult to detect diastolic or dicrotic notch for old or unhealthy individuals. In time series, we considered such signals to select the competitive features that could have been ignored and can improve the estimation accuracy. Thus, this factor leads to improving the performance using signal sequential analysis with temporal attention-based architecture.

Previously, we have conducted several baseline experiments using traditional and deep machine learning algorithms [18]. Their performances illustrate that linear model performance was lower than the neural non-linear machine

learning models. Thus, this can provide strong evidence that the linear and traditional machine learning models are unable to capture the relationship between PPG signals and blood pressure estimation. LSTM and GRU performed better than the traditional machine learning models. Further, we used an attention mechanism with a Bi-LSTM encoder and decoder that facilitates the model to extract the most competitive features in each time step. The selected features contain more relevant information related to blood pressure estimation. The proposed model achieved higher performance with optimal feature selection compared to the Bi-LSTM machine learning model using time-domain features; the error was reduced by 40%, 80% using the MIMIC dataset and 30% and 35% using the PPG-BP dataset for SBP and DBP estimation, respectively. Moreover, the proposed model outperformed GRU performance using time-domain features compared to optimal feature selection by 48% using the MIMIC dataset and 7% using the PPG-BP dataset error reduced for systolic blood pressure estimation.

The proposed model outperformed the baseline models using the MAE and SD metrics for both SBP and DBP. The temporal attention mechanism with the bi-LSTM encoder and decoder model for signal sequential analysis appears highly promising for continuous and cuffless blood pressure estimation. The results are highlighted in Table 3. These were achieved using one Bi-LSTM encoder layer with the attention layer, followed by decoder LSTM layers. The MAE and SD for optimal feature extraction with the temporal attention mechanism model are 2.31 ± 3.16 mmHg for SBP and 1.13 ± 2.32 mmHg for DBP using the MIMIC II dataset and 3.43 ± 3.08 mmHg for SBP and 3.29 ± 2.19 mmHg for DBP using PPG-BP dataset. Table 3 provides a comparison between the baseline experimental results with a proposed sequence-to-sequence bidirectional LSTM with temporal attention model for optimal feature selection to estimate SBP and DBP.

Furthermore, we provide a comparison of proposed research results with the standard protocols AMMI and BHS. The results were evaluated using two matrices, MAE and SD. In order to meet the AAMI standards, the evaluation must be performed on more than 85 subjects' data and have an average prediction result error of less than 5 mmHg, and the standard deviation should be less than 8 mmHg. With regard to the AAMI standards, the proposed model results showed that the mean error is lower than 5 mmHg and the standard deviation is less than 8 mmHg for both datasets for more than 85 subjects. Table 4 provides information about AAMI standards. Another standard protocol for blood pressure estimation is the British Hypertension Society (BHS) [69], which assigns a three-level rating to measurement accuracy (A-C). This statistic quantifies the percentage of estimation for unobserved data that are

Table 3 Comparison of the proposed methodology with baseline experiments conducted using MAE (mmHg) and SD (mmHg)

Machine learning models	Feature extraction techniques	SBP (MAE $\pm$ SD)	DBP (MAE $\pm$ SD)
Bidirectional LSTM	• Time-domain features	3.87 ± 4.79 mmHg	6.82 ± 6.29 mmHg
	• Statistical Features	5.70 ± 4.87 mmHg	7.38 ± 5.51 mmHg
	• Frequency-domain features	5.42 ± 5.21 mmHg	6.17 ± 5.89 mmHg
Gated Recurrent Unit	• Time-domain features	4.49 ± 5.58 mmHg	7.10 ± 5.85 mmHg
	• Statistical Features	7.02 ± 4.84 mmHg	6.44 ± 5.50 mmHg
	• Frequency-domain features	7.80 ± 5.04 mmHg	7.92 ± 5.59 mmHg
Sequence-to-sequence Bidirectional LSTM with Temporal Attention (This work)	• Combined Features	2.31 ± 3.16 mmHg	1.13 ± 2.32 mmHg
		3.43 ± 3.08 mmHg	3.29 ± 2.19 mmHg

Table 4 AAMI standards comparison

AAMI standards	Mean absolute error	Standard deviation
MIMIC II	SBP $2.31 \leq 5$ (mmHg)	SBP $3.16 \leq 8$ (mmHg)
	DBP $1.13 \leq 5$ (mmHg)	DBP $2.32 \leq 8$ (mmHg)
PPG-BP	SBP $3.43 \leq 5$ (mmHg)	SBP $3.08 \leq 8$ (mmHg)
	DBP $3.29 \leq 5$ (mmHg)	DBP $2.19 \leq 8$ (mmHg)

Table 5 BHS standards comparison

BHS standards		<5mmHg	<10mmHg	<15 mmHg
Grade A		60%	85%	95%
Grade B		50%	65%	90%
Grade C		40%	75%	85%
Our results				
MIMIC	SBP	74.2%	92.1%	97.2%
	DBP	85.2%	95.1%	97.5%
PPG-BP	SBP	72.6%	91.9%	95.3%
	DBP	80.1%	93.2%	96.6%

between a particular range of 5, 10, and 15 mmHg of the goal values. The proposed model achieved a Grade A for both SBP and DBP based on this criterion, which can be seen in Table 5. A histogram distribution plot presents the estimation error for systolic and diastolic blood pressure for optimal feature selection. It can be seen from Fig. 5 that the estimation error is mainly concentrated around zero. More specifically, the graph's estimation error lies between -10 and 10 for both systolic and diastolic blood pressure. The pointed shape of the histogram graph for both SBP and DBP signifies the small estimation error achieved by the proposed attention-based model for optimal feature selection.

Another statistical method can be used for analysing the association concerning the estimated and target blood pressure values to analyse the estimated error significance. We used a regression plot to illustrate the relationship between the reference and estimated values provided in Fig. 6. The regression plot reveals a significant positive relationship between the independent and dependent variables. Furthermore, Pearson correlation coefficient is used to measure the correlation's strength. The R-value is 0.92 for SBP, whereas the R-value is 0.89 for DBP. The projected values strongly match the dataset's reference blood pressure values. Moreover, the bland-Altman plots have been provided for reference and estimated values of the proposed model (Fig. 7). These graphs demonstrate the data span over the systolic and diastolic values present in the dataset. The graphs demonstrate that the error range for systolic and diastolic BP is small, primarily between -10 and 10 . This demonstrates that the model has a low error and high estimation precision while estimating SBP and DBP. The findings provided evidence for answering the research questions. *How can the feature extraction and selection from PPG signals be improved for efficient accuracy?* The proposed model results showed that combining the feature extraction techniques and selecting competitive features can improve the accuracy of the machine learning algorithm in estimating blood pressure. Further, the results of this study demonstrated that by using only PPG signals, we could estimate blood pressure more efficiently and minimise the estimation error. *How can we select the most competitive features for blood pressure estimation?* In the traditional feature

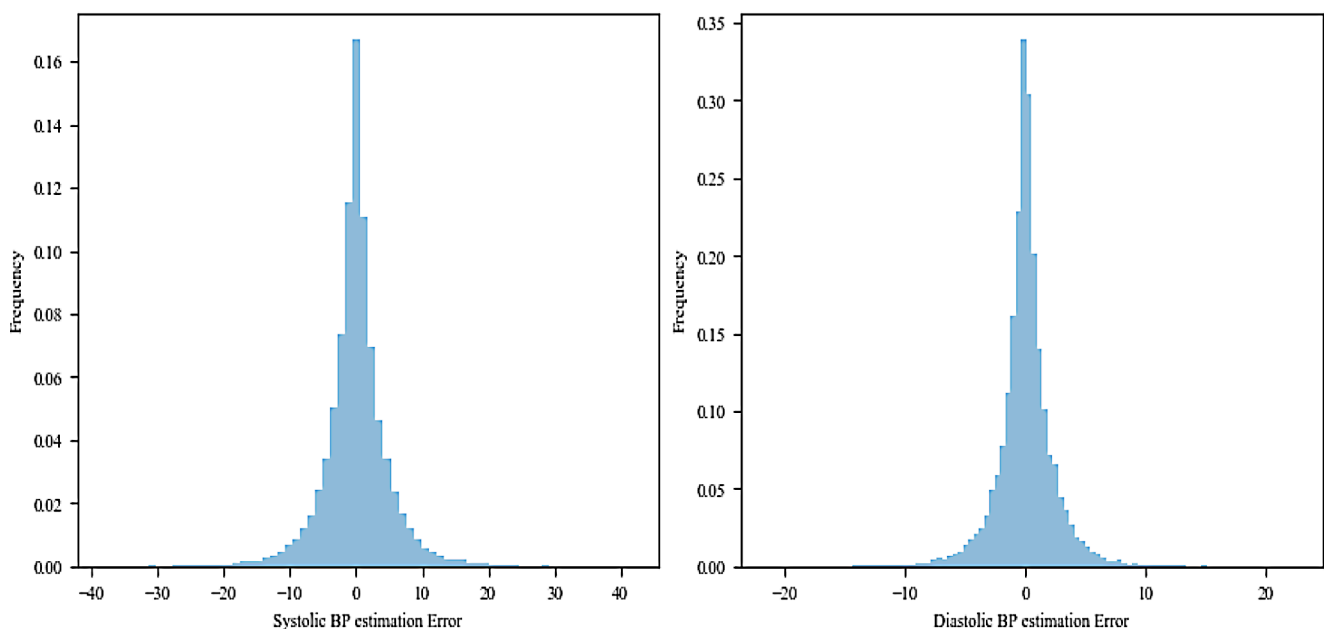


Fig. 5 Systolic and diastolic blood pressure error distribution histogram for the proposed sequence-to-sequence encoder-decoder with temporal attention mechanism for optimal feature selection

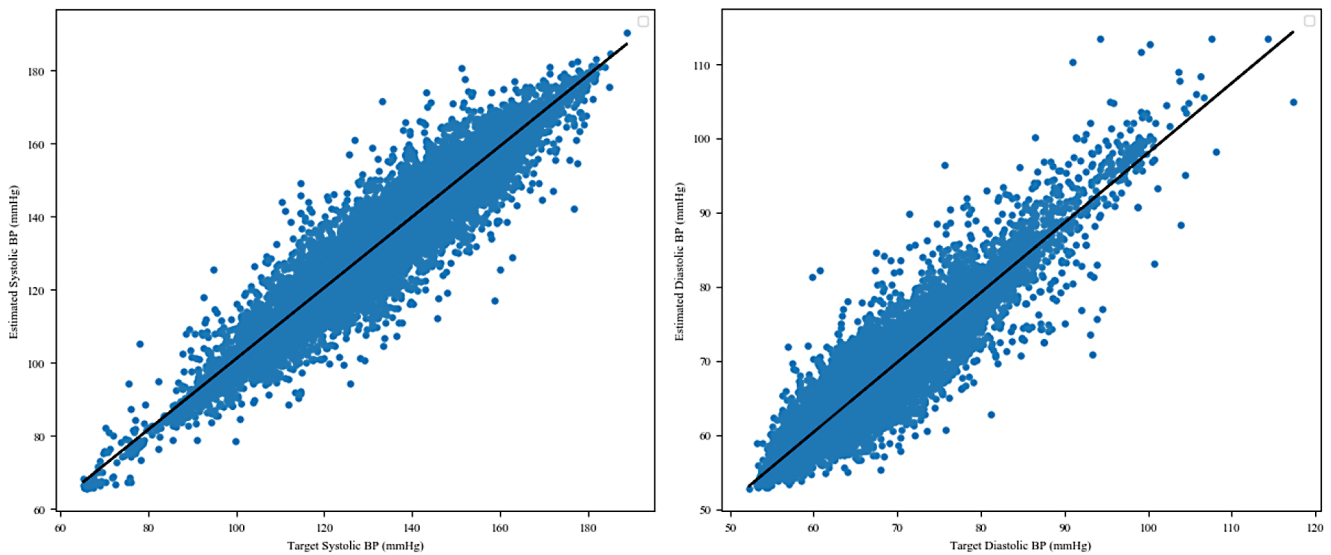


Fig. 6 The regression plot to illustrates the target and estimated SBP and DBP values for the proposed optimal feature selection model

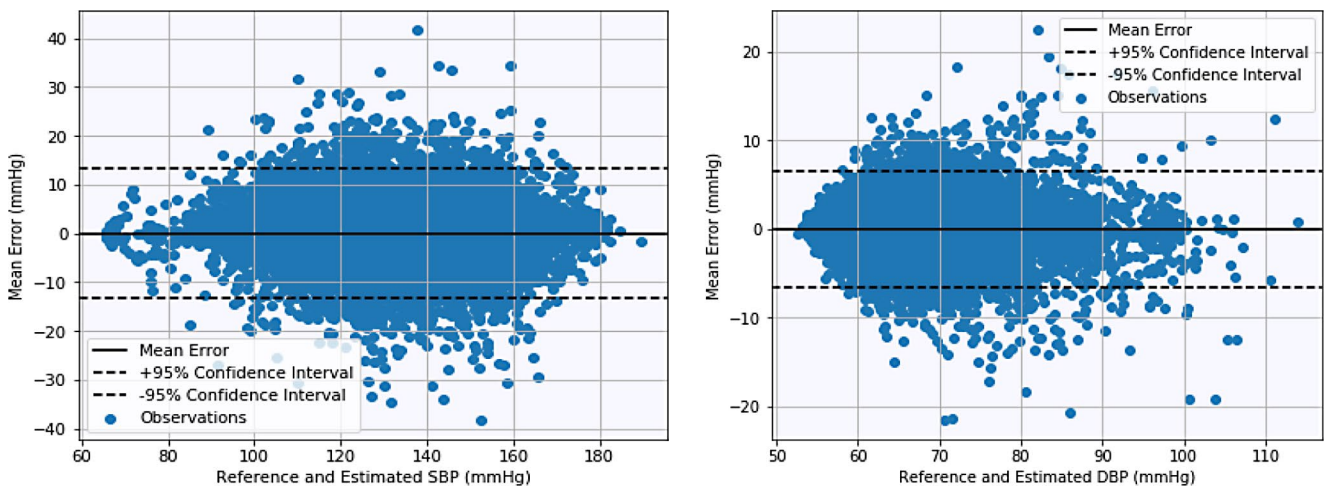


Fig. 7 Bland-Altman plots to evaluate the findings of SBP and DBP. Based on the calculated average mean error, the average estimated SBP and DBP values are displayed. This curve shows the average error differential between the anticipated and actual values

extraction method, some features may be neglected that could improve the performance if included, and some features were selected that do not contain increased prediction ability. To handle this issue and give a fair chance to all features, the attention mechanism plays a significant role in assigning and calculating weights. This research proposed an attention-based approach for continuous and cuffless blood pressure estimation from PPG signals. The novel algorithm based on the attention mechanism is proposed for feature selection, giving optimal results for accurate blood pressure estimation. The mixed complex features were provided to the Bi-LSTM encoder, decoded and attention mechanism to identify the significance of the input features for further selection. *What is the impact of feature selection on the performance of the proposed model prediction?* The experimental results showed that the performance of

blood pressure estimation was significantly enhanced by selecting the most competitive features using the attention mechanism. Further, the proposed model outperformed the state-of-the-art techniques on two datasets using only PPG signals and demonstrates the precision of this approach.

4.1 Comparison with related work

Several studies have been conducted for continuous, cuffless blood pressure estimation using PTT and PAT-based approaches using PPG and ECG signals. These approaches are computationally complex, requiring two signals' data to be processed simultaneously. Some studies conducted this research using only PPG signals that have shown promising results with low computationally complex than PTT-based methods. In Table 6, we provide a comparison summary of

Table 6 Comparison of the proposed methodology with the related work models using MAE (mmHg) and SD (mmHg)

Machine learning models	Feature extraction techniques	Error (Mean absolute error $\pm$ standard deviation)				
			SBP		DBP	
			MAE (mmHg)	SD (mmHg)	MAE (mmHg)	SD (mmHg)
AdaBoost	PAT	[47]	11.17	10.09	5.35	6.14
SVR	PTT					
GRU	Time domain	[70]	4.51	7.81	2.6	4.41
LSTM						
LSTM sequence to sequence		[19]	4.05	4.42	2.41	3.47
Bi-RNN with Attention	Time domain	[23]	2.58	3.35	1.26	1.63
LSTM based encoder	First and second derivative	[19]	4.05	5.25 (RMSE)	2.41	3.17 (RMSE)
SVR	Time domain features	[40]	2.49	7.82	1.43	4.02
Sequence-to-Sequence Bidirectional LSTM with Temporal Attention	Combined Features	This Work	2.31	3.16	1.13	2.32

our proposed deep learning model and some of the literature's relevant studies. As the data, PPG signals collection equipment, and assessment criteria are not uniform in all research, comparisons between our work and other relevant work should only be attempted as an abstract exercise. It is difficult to directly compare our study with other relevant studies due to a variety of crucial elements, including calibration for blood pressure estimation, PPG BP dataset sizes, various evaluation criteria, or distinct techniques. Furthermore, the MIMIC II dataset has seen extensive use in the academic literature, while the PPG-BP dataset is still relatively new and has seen only a small number of studies to date. In light of these considerations, we evaluate our work in terms of the MIMIC public dataset and the mean error (MAE or ME) as assessment metrics used in calibration-free research for blood pressure estimation.

Kachuee, Kiani [47] used various classical machine learning models to predict blood pressure (PAT and PPG characteristics in conjunction with PTT features), utilising traditional ML approaches. To conduct evaluations on a large dataset of test participants, the models were run using the MIMIC II dataset. Two measurement sensors were required, along with SVR and AdaBoost, which are inappropriate for modelling time-variant input features. Also, the reported MAE and SD were high. El-Hajj and Kyriacou [70] examined long-term dependencies and the performance of traditional GRU and LSTM models dealt with by these deep-learning estimate models. However, existing classical machine learning and non-recurrent models also handle time-domain data. This research took nine hundred and forty-two patients from the MIMIC II dataset. They achieved 4.51 ± 7.81 and 2.6 ± 4.41 (MAE $\pm$ SD) for SBP and DBP, respectively. Nishan, Raju [40] proposed blood pressure estimation method using PPG signals from 219 individuals. A set of frequency and time-frequency domain features were extracted from the PPG and its derivative

signals, augmented by the inclusion of age and gender as additional features for each participant. They employed support vector regression (SVR), K-nearest neighbor regression (KNN), decision tree regression (DTR), and random forest regression (RFR) models to construct the blood pressure estimation model. These regression models were trained and assessed using both the entire feature set and the selected features. Notably, the SVR model demonstrated superior performance compared to other algorithms, achieving mean absolute errors of 2.49 ± 7.82 and 1.43 ± 4.02 mmHg for SBP and DBP estimations, respectively.

Harfiya, Chang [19] introduced the LSTM sequence-to-sequence approach for continuous BP measurement using the MIMIC dataset. This method used a distinct approach and mapped the PPG signals using the LSTM translation method. The results were 4.05 ± 4.42 mmHg and 2.41 ± 3.47 mmHg for SBP and DBP estimation, respectively. The requirements of both the AAMI and the BHS were met by this study. Nevertheless, there is still a need to refine the accuracy of BP estimation. Further, El-Hajj and Kyriacou [23] used an attention mechanism with a Bi-RNN structure for BP estimation using PPG signals from two datasets that were originated from one dataset. The results showed 2.58 ± 3.35 for SBP and 1.26 ± 1.63 (MAE $\pm$ SD) for DBP estimation. However, our proposed model achieved 2.31 ± 3.16 (MAE $\pm$ SD) for SBP and 1.13 ± 2.32 (MAE $\pm$ SD) DBP estimation using PPG signals only by selection the competent features sequence-by-sequence. To summarise, our proposed model achieved high performance given the two datasets, and we utilised just PPG biosignals. The fact that BP can be estimated accurately, even when compared to the well-established PTT method, in the presence of just one PPG signal, illustrates the precision of this approach. Literature studies used traditional feature extraction techniques; however, we used smart method to select optimal features using temporal attention and considered

the sequential analysis of PPG signal. Our research results showed significant improvement by using the proposed model that outperformed the literature proposed methods.

5 Conclusion

This research proposed an attention-based approach for continuous and cuffless blood pressure estimation from PPG signals. The proposed research used two datasets, the PPG-BP and MIMIC II databases. This research study aims to investigate the association of PPG biosensor signals with cardiovascular assessment. The proposed algorithm, which uses an attention mechanism for feature selection, has shown notable improvements in the accuracy of predicting cardiovascular metrics like blood pressure. This research represents a significant step forward in applying machine learning to medical assessments, particularly for monitoring biosensor data. While we focused on optimizing feature selection within single-domain datasets, we acknowledge that our study does not fully address cross-dataset comparisons. This limitation is primarily due to resource constraints, but we aim to explore cross-dataset analysis in future work to enhance the model's generalizability. Additionally, the research questions were carefully crafted to align with the study's objectives, focusing on optimizing feature selection and improving predictive accuracy. Hence, this research study contributes to machine learning and medical assessment based on biosensors. In conclusion, this attention-based model shows great promise as a reliable tool for advancing cardiovascular assessments using biosensor data, providing a solid foundation for further exploration and broader applications.

Author contributions The first author's name is Sumbal and corresponding author. Sumbal Maqsood: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Resources, Data curation, Writing- Original draft preparation, Writing - Review & Editing, Visualization, Project administration. Shuxiang Xu: Writing - Review & Editing, Supervision, Project administration. Matthew Springer: Writing - Review & Editing, Supervision, Project administration. Rami Mohawesh: Review & Editing. Haythem Bany Salameh: Review & Editing. Shivangi Gheewala: Editing.

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Data availability No datasets were generated or analysed during the current study.

Declarations

Competing interests The authors declare no competing interests.

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